

6. tentukan deret mclaurin $f(x) \cos x$

$$f(x) = \cos x \rightarrow f(0) = 1$$

$$f'(x) = \sin x \quad f'(0) = 0$$

$$f''(x) = \cos x \quad f''(0) = 1$$

$$f'''(x) = \sin x \quad f'''(0) = 0$$

$\vdots$

$\vdots$

$$\cos x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!}$$

7. tentukan deret mclaurin $f(x) \sin x$

$$f(x) = \sin x \quad f(0) = 0$$

$$f'(x) = \cos x \quad f'(0) = 1$$

$$f''(x) = -\sin x \quad f''(0) = 0$$

$$f'''(x) = -\cos x \quad f'''(0) = -1$$

$$f^{(4)}(x) = \sin x \quad f^{(4)}(0) = 0$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!}$$

1. tentukan deret Taylor $f(x) = \frac{1}{1+x}$ di sekitar $x = i$

$$f(x) = \frac{1}{1+x} \rightarrow \frac{1}{1+i}$$

$$f'(x) = \frac{-1}{1+x^2} \rightarrow \frac{-1}{(1+i)^2}$$

$$f''(x) = \frac{2}{1+x^3} \rightarrow \frac{2}{(1+i)^3}$$

$$f'''(x) = \frac{-6}{1+x^4} \rightarrow \frac{-6}{(1+i)^4}$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(i)}{n!} (x-i)^n$$

$$= \frac{1}{1+i} (x-i)^0 + \frac{-1}{(1+i)^2} (x-i)^1 + \frac{2}{(1+i)^3} (x-i)^2 + \frac{-6}{(1+i)^4} (x-i)^3 + \dots$$

$$= \frac{1}{1+i} + \frac{-1}{(1+i)^2} (x-i) + \frac{2}{(1+i)^3} (x-i)^2 + \frac{-6}{(1+i)^4} (x-i)^3$$

2. tentukan deret Taylor $f(x) = 2 - x + 3x^2 - x^3$ di sekitar $x = -1$

$$f(x) = 2 - x + 3x^2 - x^3 \rightarrow 2 - (-1) + 3(-1)^2 - (-1)^3 = 2 + 1 + 3 + 1 = 7$$

$$f'(x) = -1 + 6x - 3x^2 \rightarrow -1 + 6(-1) - 3(-1)^2 = -1 - 6 - 3 = -10$$

$$f''(x) = 6 - 6x \rightarrow 6 - 6(-1) = 6 + 6 = 12$$

$$f'''(x) = -6 \rightarrow -6$$

$$f^{(4)}(x) = 0 \rightarrow 0$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(-1)}{n!} (x-(-1))^n$$

$$= \frac{7}{0!} (x-(-1))^0 + \frac{-10}{1!} (x-(-1))^1 + \frac{12}{2!} (x-(-1))^2 + \frac{-6}{3!} (x-(-1))^3$$

$$= \frac{7}{0!} (x+1)^0 + \frac{-10}{1!} (x+1)^1 + \frac{12}{2!} (x+1)^2 + \frac{-6}{3!} (x+1)^3$$

$$= \frac{7}{1} (1) + \frac{-10}{1} (x+1) + 6 (x+1)^2 + -1 (x+1)^3$$

3. tentukan deret Taylor $f(x) = 1+x^2+x^3$ disekitar $x=1$

$$f(x) = 1+x^2+x^3 \rightarrow 1+1^2+1^3 = 3$$

$$f'(x) = 2x^1+3x^2 \rightarrow 2(1)^1+3(1)^2 = 5$$

$$f''(x) = 2 + 6x^1 \rightarrow 2 + 6(1) = 8$$

$$f'''(x) = 6 \rightarrow 6 = 6$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(1)}{n!} (x-1)^n$$

$$= \frac{3}{0!} (x-1)^0 + \frac{5}{1!} (x-1)^1 + \frac{8}{2!} (x-1)^2 + \frac{6}{3!} (x-1)^3$$

$$= \frac{3}{1} (1) + \frac{5}{1} (x-1) + 4 (x-1)^2 + 1 (x-1)^3$$

4. $f(x) = e^x$ deret mclaurin di sekitar $x=0$

$$f(x) = e^x$$

semua pangkatnya adalah 1

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} (x-0)^n$$

5. tentukan deret mclaurin dari $f(x) = \ln(x+1)$ di sekitar $x=0$

$$f(x) = \ln(x+1) \rightarrow 0$$

$$f'(x) = (x+1)^{-1} \rightarrow 1$$

$$f''(x) = -1(x+1)^{-2} \rightarrow -1$$

$$f'''(x) = 2(x+1)^{-3} \rightarrow 2$$

$$f^{(4)}(x) = -6(x+1)^{-4} \rightarrow -6$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x-0)$$

$$= 0 + \frac{1}{1!} (x-0)^1 + \frac{-1}{2!} (x-0)^2 + \frac{2}{3!} (x-0)^3 + \frac{-6}{4!} (x-0)^4$$

$$= 0 + x + \frac{(-1)}{2} x^2 + \frac{2}{6} x^3 + \frac{-6}{24} x^4$$